RAMRUP SATPATI



Data Science & Machine Learning Engineer | ENFJ-A Protagonist

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PROFESSIONAL SUMMARY

Degree-level BS student in Data Science & Applications at **IIT Madras** (Roll: 24f3004027 | **TOPPER - Top Course Grade - Diploma**). Technically grounded in Machine Learning, Deep Learning, Generative AI systems, and Operations Research mathematical optimization models. As an **ENFJ-A (Protagonist)**, blends analytical rigor with empathetic collaboration, systems thinking, and structured cross-functional communication to lead high-impact technical initiatives bridging algorithmic intelligence with human-centered applications.

EDUCATION

Indian Institute of Technology Madras (IIT Madras) — **2024 – Present**
Bachelor of Science (BS) in Data Science and Applications (Degree Level) — **Chennai, India**

- **Academic Standing & GPA: CGPA: 8.8 / 10.0** (Degree Level) • Awarded **'TOPPER - Top Course Grade - Diploma'** Honor Badge.
- **Foundational & Diploma Certifications:** Completed all theoretical components for dual Diplomas in Programming and Data Science.
- **Core Coursework:** Machine Learning Foundations & Practice (MLF/MLP), Deep Learning & GenAI (Viva: 45/50, 90%), Business Data Management (BDM), Modern App Dev I & II (MAD 1/2), Tools in Data Science (TDS), Systems & Usable Security (Elite Distinction 81%).

TECHNICAL & LEADERSHIP SKILLS

- **Languages & Databases:** Python (Expert), SQL (PostgreSQL, SQLite), JavaScript, Bash/Shell, HTML5/CSS3.
- **Machine Learning & AI:** PyTorch, TensorFlow, Scikit-Learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, GenAI & Transformer Models.
- **Optimization & Methods:** Constrained Linear Programming (Simplex / SciPy), Customer Lifetime Value (CLV), Econometrics, EDA.
- **Web & Tools:** Flask, Vue.js, Streamlit, RESTful APIs, Git/GitHub, Docker, Linux/macOS, Jupyter Notebooks.
- **Leadership & Personal Dynamics (ENFJ-A):** Empathetic leadership, proactive psychological safety, systems thinking, milestone tracking.

FEATURED PROJECTS & REPOSITORIES

Sales and Service Mix Optimization – Primary Data Analysis (BDM Capstone) — **Oct 2026**
https://github.com/24f3004027/Business_Data_Management

- Administered a primary field survey across 220 retail/commercial clients evaluating after-sales service (AMC) attachment economics.
- Formulated and solved a constrained Simplex Linear Programming (LP) model balancing technician labor and working capital limits.
- Discovered optimal mix (74 sales units, 132 AMC contracts/mo), securing a **+28.4% profit increase** (INR 1.226M/mo) and a 3.0x return on technician payroll.

Deep Learning & Generative AI Systems — **2026**
https://github.com/24f3004027/Deep_Learning_Project

- Implemented end-to-end deep learning pipelines utilizing CNNs, recurrent sequence models, and transformer representations in PyTorch/TensorFlow.
- Applied transfer learning and latent space representations; scored **45 / 50 (90%) in oral viva defense** and 10/10 on the technical report.

Placement Portal Application V2 (MAD 2 Capstone) — **2025 – 2026**
https://github.com/24f3004027/Placement_Portal_Application_V2

- Architected an enterprise placement management platform supporting Role-Based Access Control (RBAC) for Admins, Corporate Recruiters, and Candidates.
- Built RESTful APIs with Flask, automated background job queues (Celery/Redis), and designed a reactive Vue.js interface.

Vehicle Parking Management System (MAD 1 Capstone) — **2025**
https://github.com/24f3004027/vehicle-parking-appv1

- Engineered an automated multi-slot vehicle parking application with automated ticket timestamps, dynamic rate billing, and SQLite database storage.
- Implemented server-side validation and session security in Python/Flask with clean responsive UI.

Hospital Management System – HMS (MAD 1 Improvement Project) — **2025**
https://github.com/24f3004027/HMS-MAD-Projectv1

- Constructed a comprehensive clinical records and appointment scheduling system connecting doctors, patients, and administrators.
- Designed relational database schemas, automated appointment slot booking, and generated diagnostic summaries.

Predictive Machine Learning & Classification Pipeline (MLP) — **2025**
https://github.com/24f3004027/Machine_Learning_Practice_Project

- Developed end-to-end supervised machine learning models utilizing Scikit-learn and XGBoost for classification and regression tasks.
- Executed feature transformation, cross-validated grid search hyperparameter tuning, and model explainability metrics.

Python Projects Monorepo **2024 – 2026**
https://github.com/RSNPIIT/Python-Projects
- Consolidated repository of modular algorithmic tools, data structures, automation utilities, and numerical scripts in Python.

EXPERIENCE & LEADERSHIP INITIATIVES

Open Source Contributor & Technical Project Lead **2024 – Present**
Independent Data Science & Software Developer | GitHub: RSNPIIT **Remote**
- Architected reproducible data science and optimization repositories; authored clear documentation and analytical codebooks.
- Applied ENFJ-A strengths to foster psychological safety, active team consensus, and structured milestone tracking in technical projects.

CERTIFICATIONS & LANGUAGES

- **Certifications:** Top Course Grade - Diploma Honor (IIT Madras), Systems and Usable Security (Elite Distinction 81%), Completed Foundational & Dual Diploma coursework.
- **Languages:** English (Professional Working Proficiency), Hindi (Native), Bengali.